

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Civil Engineering
Mid Semester Examination, Jan- June 2023

B. Tech: Semester-4th (CED)
Course Title: Water Supply Engineering and Pollution Control
Maximum Time: 2 hours

Course Code: CE43110
Max. Marks: 30

Instruction:

1. Attempt **All** questions. Assume any suitable data, if necessary.
2. The Marks, Course Outcome (CO) and Bloom's Level (BL) related to questions are mentioned on the right-hand side margin. Students need not to worry about the CO and BL, they have to write answers of the given question only.

Q. No	Questions	Marks	CO	BL														
1.	Derive an expression for drawdown (d) developed in steady state condition during yield 'Q' from a well of radius 'r' in an unconfined aquifer having Coefficient of permeability 'K'. The radius of circle of influence is R. The water table is H unit above from the lowest level of unconfined aquifer.	6	CO3	III														
2.	Two parallel rivers R ₁ & R ₂ are separated by 3 Km wide confined aquifer. This aquifer system consist 10 m thick (K= 10 m/day). The confined aquifer base is horizontal impervious layer. The ground level is 30 m above the top of confined aquifer. The water level of river R ₁ & R ₂ are 35 m and 15 m above the horizontal impervious base. Estimate the seepage discharge per unit length of the river from river R ₁ to R ₂ . (Hints: Use Darcy's law)	6	CO4	II														
3.	(a) List down various types of water demand for planning of water supply scheme. (b) A growing town has population as given by the Census Department is as under: Census Year: 1971 1981 1991 2001 2011 Population: 250000 480500 550300 638600 695200 In this town it has been decided to provide water at the rate of 200 lpcd. Estimate the domestic water requirement of this town in the year 2041, using Incremental Increase Method.	1 5	CO1 CO3	I III														
4.	(a) Explain the River Intake with the help of a neat sketch. What points should be kept in mind while selecting a site for the intake works? (b) Design a Bellmouth Strainer for the direct intake well system for a flow of 5.5 MLD. The highest water level is 9 m above the lake bed. Ground level is 10 m above the lake bed.	2 4	CO1 CO4	I IV														
5.	(a) Explain various factors considering while deciding Design Period. (b) Design the size of service reservoir (by using analytical method) required to supply the demand as shown in the following table if the inflow of water to the reservoir is maintained at a uniform rate throughout 24 hrs.	1 5	CO2 CO4	II IV														
	<table><tr><td>Time</td><td>0 - 4</td><td>4 - 8</td><td>8 - 12</td><td>12 - 16</td><td>16 - 20</td><td>20 - 24</td></tr><tr><td>Demand in million litres</td><td>0.48</td><td>0.87</td><td>1.33</td><td>1.0</td><td>0.82</td><td>0.54</td></tr></table>	Time	0 - 4	4 - 8	8 - 12	12 - 16	16 - 20	20 - 24	Demand in million litres	0.48	0.87	1.33	1.0	0.82	0.54			
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